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|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(a)(i)  | <b>X</b> neutrophil ;<br><b>Y</b> monocyte ; <b>A</b> lymphocyte <b>I</b> B– / T–                                                                                                                            | <b>2</b> |
| 3(a)(ii) | (magnification) = $\frac{\text{image / line Z, length / width}}{\text{actual width}}$ ; <b>A</b> (M) = I / A<br><b>A</b> magnification triangle<br>(x) 1500 ; <b>A</b> 1375 (for 11 mm) / 1438 (for 11.5 mm) | <b>2</b> |

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks |
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| 3(b)(i)  | <p>any <b>four</b> from:</p> <ol style="list-style-type: none"> <li>1 (carbonic anhydrase) catalyses the reaction between <math>\text{CO}_2</math> and <math>\text{H}_2\text{O}</math> ; AW</li> <li>2 to form, carbonic acid / <math>\text{H}_2\text{CO}_3</math> ;</li> <li>3 <math>\text{H}^+</math> forms (together with <math>\text{HCO}_3^-</math> from dissociation of <math>\text{H}_2\text{CO}_3</math>) ;</li> <li>4 <math>\text{H}^+</math> binds to haemoglobin to form haemoglobinic acid ;</li> <li>5 displacing / releasing, oxygen from haemoglobin ;</li> <li>6 allosteric effect / change in tertiary structure / AW, in (oxy)haemoglobin (causes release of oxygen) ;</li> </ol> | 4     |
| 3(b)(ii) | <p>curve below the original line ;</p> <p>curve, merging / will merge, with the original line after it begins to plateau ;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2     |

| Question | Answer | Marks |
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